

PlayIQ Module 6: Self-Testing and Mistake Bank

Your mission: turn wrong answers into your fastest study map

What if every mistake could tell you exactly what to learn next?

Most students hide mistakes, erase them, or reread the whole chapter. In this module, you will do something more powerful: test what you can produce without looking, investigate why an answer missed, and save the useful clue in a Mistake Bank.

By the end of this module, I can:

- explain why retrieval reveals more than passive rereading;
- classify a mistake without treating it as a personal failure;
- compare one-at-a-time and full-set quiz formats;
- correct an answer and prove the correction with a new question;
- create a realistic spaced-review plan without pretending a reminder exists;
- give Orion a testable quiz-and-feedback setting;
- add evidence-backed mistake and review rules to my future tutor Project.

ORION: "A wrong answer is not a verdict. It is a message from the exact

edge of what you know. If we read that message correctly, it becomes a map."

Part 0: The 30-second confidence hook

Without looking back at Module 5, answer quickly:

What were the three supporting parts on your Understanding Card called?

How confident are you?

- ☐ very confident
- ☐ somewhat confident
- ☐ guessing

Now check Module 5.

Result:

- ☐ correct
- ☐ partly correct
- ☐ not yet

There is no penalty. In less than a minute, you discovered two pieces of useful information: what you could retrieve and whether your confidence matched the result. That is more useful than merely feeling that the page looked familiar.

Part 1: Rereading can feel easier than remembering

When the answer is visible, your brain can recognize it. Recognition feels familiar, but a test usually asks you to produce, explain, select, or use the idea without the answer sitting in front of you.

Self-testing closes that gap:

1. Hide the source.
2. Try to retrieve or use the idea.
3. Check against an approved source.
4. Investigate the mistake.
5. Correct it in your own words.
6. Try a new question that tests the same missing skill.

Orion conversation: coach, not answer machine

STUDENT: "I got it wrong. Just tell me the answer."

ORION: "I will help, but first let us use the mistake. Was the problem a missing fact, an unclear word, a skipped step, a rushed error, difficulty using the idea in a new situation, or confidence that did not match the answer?"

STUDENT: "I knew the rule, but I applied it to the wrong number."

ORION: "That may be a step or attention error rather than missing knowledge. Show me the line where you chose the number. Then I will give a hint and a similar question so you can prove the correction."

Why this matters: the same wrong answer can need very different repairs. More rereading will not fix every kind of mistake.

Part 2: The six Mistake Bank categories

Categories are working hypotheses. You can revise one after looking more closely.

1. Knowledge gap

You did not yet know or remember an important fact, rule, or concept.

Example: You cannot explain the purpose of mitochondria because the function was never learned clearly.

Useful repair: return to the core explanation, then retrieve it again.

2. Word gap

An important word, symbol, or instruction was misunderstood.

Example: You answer only one side of a question because you misread contrast.

Useful repair: define the term in context and compare an example with a non-example.

3. Step error

You knew the idea but skipped, reversed, or misapplied a step.

Example: You forgot to distribute a negative sign through both terms.

Useful repair: identify the decision point, explain why the step is needed, then try a matched example.

4. Careless or attention error

Your understanding may be present, but the response contains a copying, reading, sign, unit, or checking mistake.

Example: You solve correctly but copy 42 as 24 in the final line.

Useful repair: create a final check that targets the exact repeated risk.

Do not use "careless" as an insult. It describes the event, not the person.

5. Transfer error

You understand the idea in a familiar example but do not yet recognize how to use it in a different situation.

Example: You can identify a metaphor in a practice sentence but miss it inside a poem.

Useful repair: compare the familiar and new cases, identify what stayed the same, and practise one more varied example.

6. Confidence error

Your confidence and your demonstrated answer did not match.

Examples:

- You were certain about an incorrect answer.
- You had a correct explanation but changed it because you did not trust it.

Useful repair: record confidence before checking and learn which clues deserve trust.

Quick classification practice

Choose the most likely category and explain why.

1. Maya knows the formula but changes the denominator when only the numerator should change.

Hypothesis: _____ Why: _____

2. Leo says he remembers the chapter after rereading, but cannot name the chapter's central claim when the book is closed.

Hypothesis: _____ Why: _____

3. Sam correctly solves the practice example but cannot use the same principle in a word problem.

Hypothesis: _____ Why: _____

4. Noor understands the evidence but thinks *justify* means "give the answer" rather than "support it with reasons or evidence."

Hypothesis: _____ Why: _____

ORION: "The category is useful only if it changes the next practice move.

We are not collecting mistakes to judge you. We are collecting clues so you

do not waste practice on the wrong problem."

Part 3: Create your Mistake Bank

Choose one recent wrong or incomplete answer from school-approved material. If you do not have one, ask Orion for a short practice question on a topic you are

learning. Attempt it before receiving feedback.

Mistake Bank entry

Date: _____

Subject and topic: _____

Question or task, summarized in my own words:

My first answer:

My confidence before checking: ☐ high ☐ medium ☐ low

Approved source or answer used to check:

What was correct in my answer?

What was missing or incorrect?

First category hypothesis:

- ☐ knowledge gap
- ☐ word gap
- ☐ step error
- ☐ careless or attention error
- ☐ transfer error
- ☐ confidence error
- ☐ not yet known

Evidence for the category:

The repair loop

Ask Orion:

"Act as my learning coach. Here is the question, my answer, and the approved

source or teacher feedback. Help me classify the mistake. Ask me one question

before giving a hint. Do not rewrite my answer. After I correct it, give me a

new question that tests the same skill in a different example."

Orion's diagnostic question:

My answer:

Hint received, if needed:

My corrected explanation or method:

New transfer question:

My answer:

Transfer result: ☐ correct ☐ partly correct ☐ not yet

Did the original category still fit? ☐ yes ☐ no ☐ unsure

Revised category, if needed: _____

Why this activity matters

A correction is not proven just because the right answer is visible. The new question checks whether you can use the repaired idea yourself.

Part 4: Learn How You Learn discovery

1. What is your first emotional reaction when an answer is wrong or incomplete?

2. What kind of feedback helps you stay in control and try again?

3. What kind of feedback makes you stop thinking or feel overwhelmed?

4. Do you usually notice mistakes during the work, only after checking, or only when someone points them out? Give an example.

5. When would immediate feedback help you? When might finishing a short set before feedback be better?

6. What should your future personal tutor say after a wrong answer so the mistake feels useful but is not ignored?

These responses can inform Orion's tone and feedback sequence, but the active comparison below provides stronger evidence about what worked on this task.

Part 5: Active comparison—two quiz formats

Compare two formats using closely matched questions from the same topic. Ask

Orion to keep difficulty similar and to avoid showing answers before you try.

Format A: One question at a time

Prompt:

"Quiz me on [topic] with five questions, one at a time. Wait for each answer.

Ask for my confidence before checking. If I am wrong, ask one diagnostic

question, give one hint if needed, and give a matched follow-up question. Do

not continue until I attempt the repair."

Record the run:

Question	First result	Confidence matched?	Hints	Follow-up result
1	Correct / Partial / Not yet	Yes / No / Unsure	—	Correct / Partial / Not yet
2	Correct / Partial / Not yet	Yes / No / Unsure	—	Correct / Partial / Not yet
3	Correct / Partial / Not yet	Yes / No / Unsure	—	Correct / Partial / Not yet
4	Correct / Partial / Not yet	Yes / No / Unsure	—	Correct / Partial / Not yet
5	Correct / Partial / Not yet	Yes / No / Unsure	—	Correct / Partial / Not yet

Focus during this format (1–5): ____

Frustration after this format (1–5): ____

What immediate feedback helped or interrupted:

Format B: Full set, then feedback

Prompt:

"Give me five new questions on [same topic] at similar difficulty. Do not

show answers or feedback until I submit all five. Ask me to mark confidence

beside each answer. Then help me classify the mistakes and choose one repair

at a time."

Record the run:

Question	First result	Confidence matched?	Mistake category, if any	Repair result
1	Correct / Partial / Not yet	Yes / No / Unsure	_____	Correct / Partial / Not yet
2	Correct / Partial / Not yet	Yes / No / Unsure	_____	Correct / Partial / Not yet
3	Correct / Partial / Not yet	Yes / No / Unsure	_____	Correct / Partial / Not yet
4	Correct / Partial / Not yet	Yes / No / Unsure	_____	Correct / Partial / Not yet
5	Correct / Partial / Not yet	Yes / No / Unsure	_____	Correct / Partial / Not yet

Focus during this format (1–5): _____

Frustration after this format (1–5): _____

What uninterrupted answering helped or made harder:

Fair comparison checklist

- ☐ Both quizzes used the same topic.
- ☐ The questions were similar in difficulty, as far as I could judge.
- ☐ Both had five questions.
- ☐ I answered before seeing feedback.
- ☐ I marked partial answers honestly.
- ☐ I used results as local evidence, not an official grade or PDI score.

Part 6: Evidence interpretation

Evidence	One-at-a-time	Full set
First answers correct	___ / 5	___ / 5
Partial answers	___	___
Not-yet answers	___	___
Repairs completed	___	___
Focus (1–5)	___	___
Frustration (1–5)	___	___
Confidence mismatches noticed	___	___

Which format helped you notice and repair mistakes more clearly?

Which format helped you maintain focus?

Did one format produce better first answers while the other produced better

repairs?

What evidence is strongest? What evidence is still uncertain?

What else may have affected the comparison?

Interpret without a fixed label

Do not conclude, "I can only be tested one question at a time."

Prefer:

"For this topic and session, _____ supported _____. I want Orion to

try that format first when _____, and use _____ when I need to

practise working through a set without interruption."

My evidence-based interpretation:

No PDI score is calculated in this module. Your answer results, repair results, confidence observations, and review plan are useful evidence on their own.

They must not be converted into an unapproved composite formula or an invented parent-dashboard claim.

Part 7: Build a spaced-review plan

One successful correction may fade. Spaced review means returning after time has passed so retrieval requires effort again.

A starting rhythm you may test is:

- **Review 1:** later today or after a meaningful break;
- **Review 2:** the next day;
- **Review 3:** about three days later;
- **Review 4:** about one week later.

This is a starting plan, not a guaranteed memory formula. School deadlines, topic difficulty, your schedule, and your results may require changes.

Choose how you will remember

Use only a method actually available and approved by your parent:

- ☐ paper planner
- ☐ parent-approved calendar entry
- ☐ school agenda
- ☐ phone reminder approved by parent
- ☐ checklist kept with the Mistake Bank
- ☐ another real method: _____

Do not assume PlayIQ, Orion, a dashboard, or an AI platform will automatically send an alert unless that behavior is visibly available and verified in the tool you are using.

My review plan

Review	Planned date/time	Retrieval question	Result after completion
1	_____	_____	Correct / Partial / Not yet / Pending
2	_____	_____	Correct / Partial / Not yet / Pending

3			Correct / Partial / Not yet / Pending
4			Correct / Partial / Not yet / Pending

How will I decide whether to continue, change, or retire this review item?

Example rule:

"If I retrieve and apply the idea correctly on two separated reviews, move it

to a later check. If I miss it, diagnose the new mistake and practise a

matched example before scheduling the next review."

Part 8: Learning Supercharger Blueprint update

This setting tells your future tutor how to begin quizzes and respond to mistakes. It is a testable instruction, not a claim that one format always wins.

Proposed tutor settings

quiz-start = [one-at-a-time | full-set | depends-on-purpose | not-yet-known]

best-use-condition =

after-wrong-answer =

hint-policy = "Ask me one diagnostic question before giving a hint."

proof-policy = "After a correction, give me a new matched question."

fallback-format =

Write the complete rule in your own words:

"When I am practising _____, begin with _____.

After a wrong or incomplete answer, _____. Do not reveal the

answer before I try. After I correct it, _____."

Rule status:

- [] `experiment\_result`
- [] `student\_reported`
- [] `pending\_validation`

Part 9: Structured Knowledge File updates

Keep the tutor rule separate from individual mistake records. This helps your future personal tutor understand both **how to quiz you** and ****what needs more practice****.

A. Learning rule record

Append to Learning_Rules.md:

```
- rule_id: module_6_quiz_and_repair
module: 6
skill_tested: self_testing_and_mistake_repair
date: "[YYYY-MM-DD]"
source_activity: "one_at_a_time_vs_full_set_quiz"
topic: "[topic]"
proposed_quiz_start: "[format or not_yet_known]"
evidence:
one_at_a_time_result: "[observations]"
full_set_result: "[observations]"
repair_observation: "[what helped correction]"
confidence_observation: "[what was noticed]"
```

```

after_wrong_answer: "[student-approved response sequence]"
fallback_format: "[format or pending_validation]"
provenance: "[experiment_result|student_reported|pending_validation]"
confidence: "[low|medium|high]"
privacy: "safe_for_tutor_project"
review_condition: "Retest for a different purpose or when the setting stops helping."

```

B. Mistake Bank record

Append to Mistake_Bank.md:

```

- mistake_id: "[date-topic-short-label]"
module_added: 6
date: "[YYYY-MM-DD]"
subject: "[subject]"
topic: "[topic]"
source: "[school-approved source or practice origin]"
first_answer_summary: "[student-authored summary]"
category_hypothesis: "[knowledge|word|step|attention|transfer|confidence|unknown]"
evidence_for_category: "[observation]"
correction_in_students_words: "[correction]"
matched_question_result: "[correct|partial|not_yet]"
review_status: "[scheduled|in_progress|retired|pending]"
next_review: "[date or pending]"
provenance: "student_work_plus_checked_source"
privacy: "[safe_for_tutor_project|parent_approval_required|do_not_include]"

```

Ask Orion:

"Check these Module 6 records for unsupported claims. Separate my quiz-format

preference from the experiment result. Do not calculate a PDI or invent

progress. Identify any missing source, privacy, or pending-validation field."

One issue Orion found or one field confirmed complete:

Part 10: Proof Artifact—Quiz and Mistake Repair Card

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PLAYIQ QUIZ AND MISTAKE REPAIR CARD

=====

Student: _____ Date: _____

Topic tested: _____

QUIZ COMPARISON

One-at-a-time: correct ____ / 5 | partial ____ | focus ____ / 5

Full set: correct ____ / 5 | partial ____ | focus ____ / 5

Proposed `quiz-start`: _____

Best-use condition: _____

Fallback: _____

MISTAKE REPAIR

Category hypothesis: _____

Correction made: _____

Matched question: [correct | partial | not yet]

REVIEW

Real reminder method selected: _____

Next review date: _____

Rule status:

[experiment_result | student_reported | pending_validation]

No PDI or dashboard result claimed. [checked]

=====

Save the card, learning rule, and Mistake Bank record. These become evidence for how your personal tutor Project should quiz, correct, and schedule practice when you assemble it later in the course.

Part 11: Mastery Challenge—repair and retest

Choose one real mistake that matters to your current learning.

1. Attempt a question without looking at the answer.
2. Mark your confidence.
3. Check it against an approved source.
4. Classify the mistake as a hypothesis.

5. Ask Orion one diagnostic question and request a hint only if needed.
6. Correct the explanation or process in your own words.
7. Complete a new matched question.
8. Schedule a later retrieval using a reminder method you can actually access.

Mastery record

Topic: _____

First result: ☐ correct ☐ partial ☐ not yet

Confidence before checking: ☐ high ☐ medium ☐ low

Mistake category and evidence:

My correction:

New matched question:

My answer:

Matched result: ☐ correct ☐ partial ☐ not yet

If it is still not correct, what is my next smallest repair?

Later review date and reminder method:

Mastery check

- ☐ I tested before looking.
- ☐ I used a mistake category to choose a repair.
- ☐ I corrected the idea in my own words.

- [] I attempted a new question instead of only reading the correction.
- [] I made a real review plan without claiming an unavailable alert.
- [] I saved both knowledge-file records with provenance.

The later review may remain marked pending; never fill in a result before it happens. When it is complete, return and record the actual evidence.

ORION: "You now know how to find the edge of your knowledge and decide

what deserves another attempt. In Module 7, you will organize useful context

and study materials so your tutor can work from clear, approved knowledge

instead of guessing what you mean."

Beta tester feedback—complete only in the beta route

This section appears only after the module is complete in the beta-testing route. Standard-course students do not complete it.

1. Did the fast confidence hook make self-testing feel clear and safe?

2. Which Mistake Bank category needs a better example?

3. Where did the lesson feel too long, confusing, or repetitive?

4. Which quiz format kept you thinking most actively, and what happened?

5. Did any Orion prompt reveal too much or fail to give enough guidance?

6. What would make mistake repair feel more motivating without hiding the mistake?

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